



Assessing Iron Ore Pellet Quality Parameters for Optimized Performance Across Pellet-Based Furnace Operations

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Abstract

The increasing use of iron ore pellets as a substitute for sinter in the blast furnace is a promising strategy to reduce CO₂ emission from the steel industry. However, the diverse chemical composition and physical characteristics of commercially available pellets can significantly affect their behavior in the furnace conditions. The current study investigated the effects of key pellet parameters—total iron (T. Fe) content, Al₂O₃ and MgO content, basicity (B_2), porosity and pellet size—on physicochemical properties. Cold compressive strength (CCS), reduction degradation index (RDI), reducibility, and swelling index were evaluated for several pellet brands of inherently different composition. The results indicated that pellet performance cannot be estimated solely on the basis of chemical composition and mechanical strength. Although one pellet brand exhibited superior CCS, it showed inferior reducibility and excessive swelling, highlighting the importance of considering in-furnace behavior during pellet selection. Multiple linear regression (MLR) equations were developed as an exploratory statistical approach to estimate the relative influence of pellet characteristics with the investigated commercial dataset. Within the dataset, favorable performance was generally associated with lower MgO, moderate B_2 , and porosity near 20%. Additionally, the pellet size was found to play a critical role: larger pellets improved mechanical strength and swelling behavior, while smaller pellets enhanced reducibility and RDI performance. An intermediate pellet size ranging from 9.5 to 12 mm was identified to be optimal. The findings might provide practical guidance for pellet selection and design in high-pellet furnace operations, supporting both operational stability and emission reduction goals.

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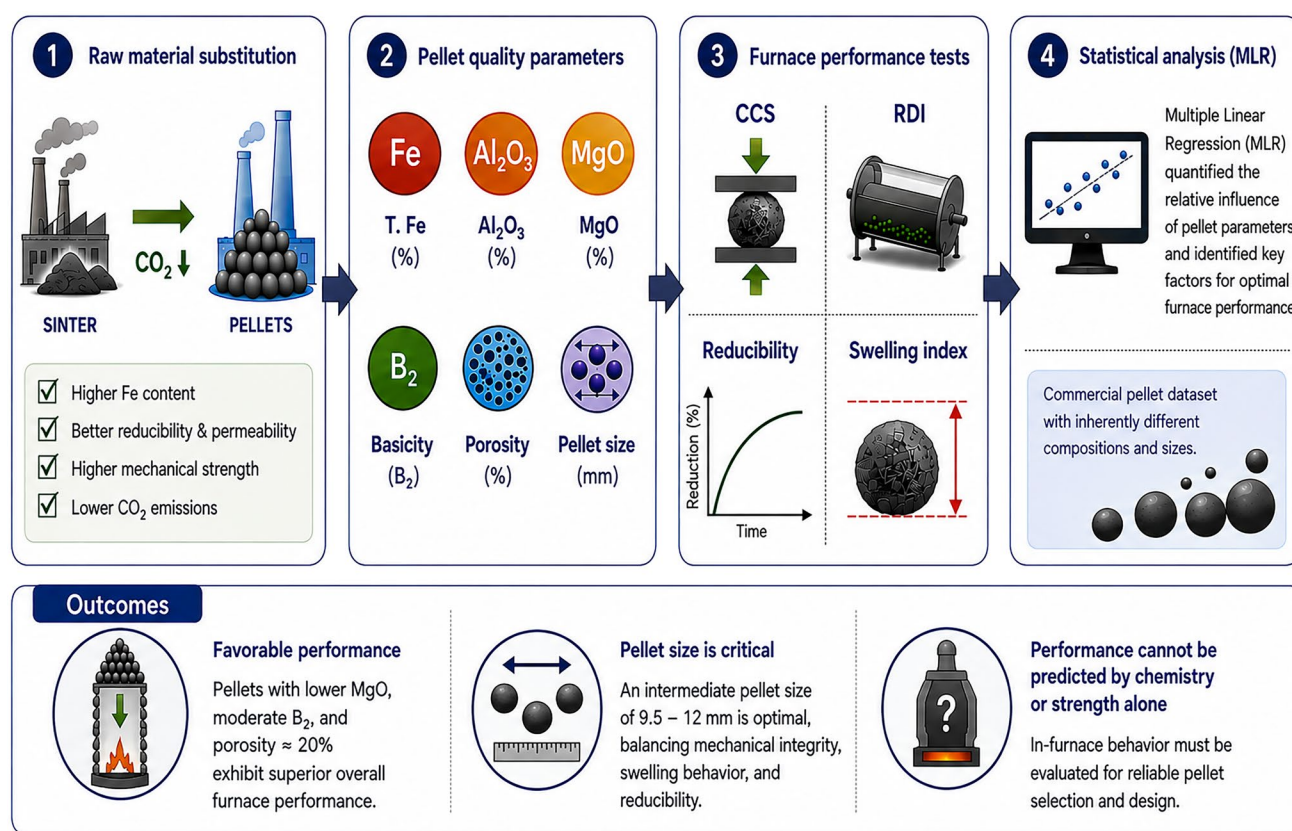
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Graphical Abstract



Keywords Iron ore pellets · Pellet quality · Physicochemical properties

Introduction

The steel industry accounts for approximately 7% of total global CO₂ emission [1, 2]. Among the various steelmaking processes, more than 80% of CO₂ emissions originate from the ironmaking stage, which includes the blast furnace, sinter plant, and coke oven [3]. The blast furnace is the largest source of CO₂ in steel plants, accounting for over 50% of total emissions, and remain as the predominant route of global steel production, representing about 70% of global output [4].

One potential strategy to mitigate the high level of emission from the blast furnace is the replacement of raw materials. Iron ore sinter is the major iron-bearing material charged into blast furnaces, especially in China, South Korea, and Japan [5]. Replacement of sinter with iron ore pellets offers several advantages, including uniform size distribution, improved reducibility and permeability, higher Fe content, and larger mechanical strength [6, 7]. These benefits can lead to the improved performance of the blast furnace including fuel efficiency.

The substitution of pellets for iron ore sinters is of great importance, which results in a significant reduction in CO₂ emissions from the agglomeration process of iron ores [7–9]. According to analyses by Lv et al. [7], the production of iron ore sinter generates approximately 4 times more greenhouse gases than that of iron ore pellets. Additionally, owing to the superior reducibility of pellets compared with ore sinter by solid fuel [7, 10], the amount of coke required for the blast furnace operation can be expected to decrease, further contributing to lower CO₂ emissions. However, as the utilization of pellets in the blast furnace is anticipated to increase, it is necessary to investigate how this shift may affect key operational characteristics such as the formation of the cohesive zone, pressure drop, and *s*-value during high-pellet operations [11, 12]. As an example, pellets generally have lower basicity than iron ore sinter. As the substitution ratio of pellets for sinter increases, the overall basicity of the iron burden decreases, which requires to take compensatory measures to maintain a similar smooth operation, especially in terms of slagging properties.

Another important factor to be considered in the high utilization of iron ore pellets in the blast furnace is the diversity of raw materials. Pellets are manufactured from different iron ore sources, which results in a wide range of chemical composition, especially in terms of total Fe, SiO_2 , Al_2O_3 , CaO, and MgO. Consequently, the parameters such as basicity and porosity will also be affected by the type of iron ore used during the pelletizing process. As a result of this diversity, the behavior of iron ore pellet will be drastically distinct, which might affect the smooth operation of blast furnace. Several studies have been already done to understand the effect of some pellet parameters such as gangue content, basicity, and porosity on the iron ore pellet properties such as TI, CCS, RDI, RI, and swelling behavior [13–17]. However, most of the previous studies used a single pellet brand as raw material, and varied the numbers of the parameters to be investigated. For example, the pellet has a fixed basicity, but CaO was added to the pellet at different content to understand the effect of basicity on the pellet properties.

As is different from the previous research, the current study evaluated the physicochemical properties of various pellet brands including mechanical strength, low-temperature degradation, reducibility, and swelling in views of the difference in chemical composition such as Fe_2O_3 , basicity, Al_2O_3 , and MgO. This investigation would tend to optimize the selection criteria for iron ore pellets to be highly used in the reactors without compromising their smooth operation. This study aims to serve as foundational data for future strategies for high-pellet operation.

Experimental

Materials Preparation

The samples used in this study were iron ore pellets of several brands originating from diverse regions in the

world, manufactured from different pelletizing companies. In total, ten brands of iron ore pellets were received directly from steel plants and used, as shown in Table 1, which summarizes the chemical composition of each brand. The samples were analyzed by wet chemical titration [18], in terms of T. Fe content, and inductively coupled plasma-atomic emission spectrometry (ICP-AES) [19], in terms of the metal oxides with exception to Fe_2O_3 . As the iron ores used to manufacture the pellets originated from different parts of the world, their chemical composition was very diverse with special focus on the gangue components such as SiO_2 , Al_2O_3 , CaO, and MgO. As a consequence, such discrepancy in the chemical composition affects other parameters such as basicity, melting behavior, and viscosity. Some important highlights should be made from Table 1. The SiO_2 content of brand H, the Al_2O_3 content of brand C, the MgO content of brands A and G, and the basicity (B_2) of brand I. The aforementioned values are the most discrepant from the averaged chemical composition of iron ore pellets. It is also important to note that the T. Fe content of all pellet brands used in this study falls into the identical range, which belongs to the higher grade of iron ore pellets compared with iron ore sinter which mostly contains a T. Fe content below 60 wt% [20].

It is important to mention again that this study did not prepare the pellets, and the samples were received already in their final stage, ready to be charged, from the major suppliers of steel plants. These ten kinds of pellets represent most of the iron ore pellets being charged into blast furnaces worldwide. Therefore, to highlight an important part that the authors are sure is essential to understand the mechanical behavior of pellets, this study did not have access to the pelletizing conditions, such as temperature, time, and atmosphere of firing steps, of each pellet, as this is considered a

Table 1 Chemical composition of the iron ore pellets used (wt%)

Pellet brands	T. Fe	Fe_2O_3	SiO_2	Al_2O_3	CaO	MgO	B_2	B_4	Theoretical oxygen that can be removed
A	65.50	93.57	2.27	0.59	2.23	0.75	0.98	1.04	28.12
B	66.50	95.00	3.09	0.64	1.29	0.06	0.42	0.36	28.55
C	64.50	92.14	2.97	1.55	2.39	0.08	0.80	0.55	27.69
D	66.90	95.57	1.58	0.34	1.72	0.32	1.09	1.06	28.72
E	65.80	94.00	2.50	0.45	2.52	0.09	1.01	0.88	28.25
F	66.30	94.71	2.26	0.59	1.76	0.20	0.78	0.69	28.47
G	66.50	95.00	2.56	0.42	0.12	2.24	0.05	0.79	28.55
H	65.62	93.44	4.46	0.40	1.09	0.31	0.24	0.29	28.08
I	67.23	95.91	1.20	0.37	1.72	0.26	1.43	1.26	28.83
J	67.79	96.72	1.25	0.38	0.96	0.33	0.76	0.79	29.07

* Fe_2O_3 was evaluated by simple assumption that all T. Fe refers to Fe^{3+}

* B_2 refers to the CaO/ SiO_2 ratio

* B_4 represents the $(\text{CaO} + \text{MgO})/(\text{SiO}_2 + \text{Al}_2\text{O}_3)$ ratio

company procedure to prepare their own products. However, this study does know that the pelletizing conditions are important factors affecting the pellet quality.

The further discussion relied mostly on the bulk chemical composition of the pellets, other than the mineral phases specially formed by the gangue components, as the major phase present in all pellet brands was found to be Fe_2O_3 . As is known, the pelletizing process contains a firing step, with the target being to oxidize any of Fe_3O_4 present in the iron ore into Fe_2O_3 when magnetite ore is used as the raw material. According to the analysis, some of the received samples (pellets B, E, G, and I) in terms of X-ray diffraction (XRD), the major distinct phase found among the analyzed pellets was only observed for brand G, which showed a slight presence of magnesioferrite ($\text{MgO} \cdot \text{Fe}_2\text{O}_3$) in their composition due to the considerable presence of MgO. This presence was highlighted by the shift in some of the Fe_2O_3 peaks to the right side, because of the distortion in the crystal structure promoted by the presence of MgO. The XRD analysis can be found in the online supplementary material, labelled as Fig. S1.

Experimental Procedure

The current research evaluated several properties of different brands of iron ore pellets. Firstly, it was attempted to measure the porosity of each pellet brand. Then, the mechanical strength in terms of cold compressive strength (CCS), reduction disintegration index (RDI), reduction degree, and swelling behavior were evaluated. The final objectives were to assess how the diverse variables of pellets, such as chemical composition and porosity, affect the major quality parameters of pellets for enhancing higher utilization in the reduction furnaces. The selected pellet size for this investigation was between 13 and 16 mm.

The porosity of iron ore pellets was evaluated in terms of image analysis by applying ImageJ software. Such a method was selected to estimate the porosity of iron ore pellet instead of the mercury porosimeter technique as the sample used for mercury porosimeter analysis is a small particle, which does not represent the total porosity of the pellet. However, the cross-sectional image analysis of the iron ore pellet analyzed the total region of the pellet with more reliable data. Figure 1 summarizes the stepwise procedure to evaluate the porosity of pellets by image analysis. From the original cross-sectional image of each pellet, the dark points are removed from the pellet area, and their ratio was evaluated to be porosity.

The cold compressive strength (CCS) was evaluated by ISO 4700 [21] in terms of loading velocity, with a modification in the amount of pellets tested. The standard requires 60 pellets, while this study analyzed the average value of about 10 pellets employing an Instron tester machine.

The reduction disintegration index (RDI) was evaluated according to ISO 4696-2 [22] in terms of temperature and gas composition of the test. Then, the reduction disintegration test was performed to evaluate the strength of pellet which were subjected to the low-temperature reduction process for simulating the upper part of the blast furnace. About 10 pellets were non-isothermally heated at a rate of $100\text{ }^\circ\text{C}/\text{min}$ from room temperature to $550\text{ }^\circ\text{C}$ in a N_2 atmosphere in a horizontal gold-image furnace. After the target temperature was reached, the gas composition was changed to 70 vol% N_2 and 30 vol% CO or H_2 , and the sample was kept at a flow rate of 1 L/min for 30 min. After the samples were cooled to room temperature, the total weight was measured. Then, the samples were tumbled at 60 rpm for 15 min. Finally, the samples were sieved in different

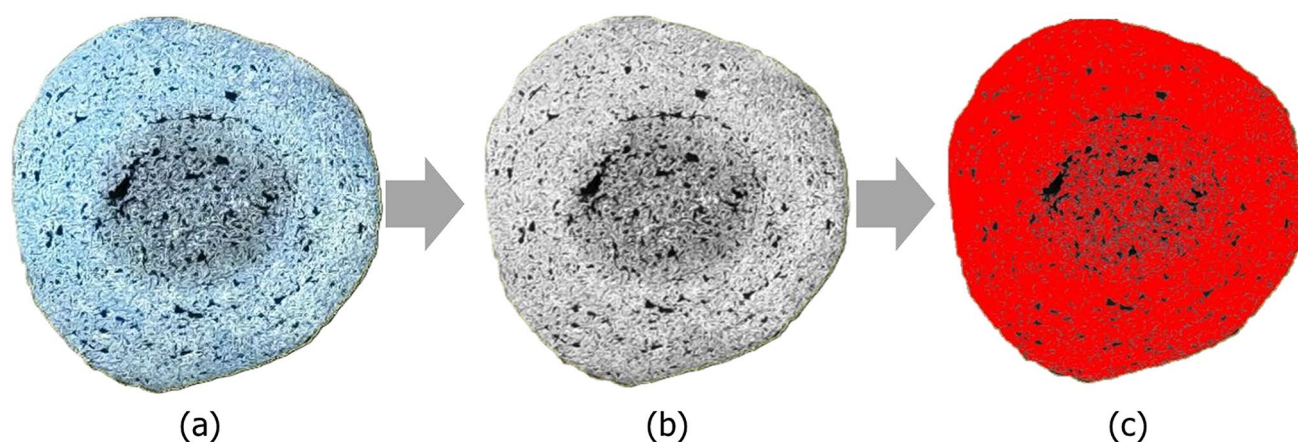


Fig. 1 Evaluation method for the porosity of iron ore pellets in terms of cross-sectional image analysis. **a** Image from a random pellet cross-section; **b** image adjusted by the brightness/contrast for an easier differentiation between the dark and grey points; and **c** red-colored

region determining the total area of the cross-section, while the dark points represent the pores present in the cross-section area. Porosity is evaluated by quantifying the fraction of these dark points in the red-colored region (dense pellet)

aperture sizes such as 3.15 and 0.5 mm. The $RDI_{+3.15\text{ mm}}$ was evaluated as the weight of samples with a particle size bigger than 3.15 mm, as represented by Eq. [1].

$$RDI_{+3.15\text{ mm}}(\%) = \frac{m_1}{m_0} \times 100 \quad (1)$$

where m_0 is the weight of the reduced sample before tumbling (g) and m_1 is the weight of samples retained in the sieves of 3.15 mm (g).

The reduction tests were carried out to measure the amount of oxygen removed from the pellets by indirect reduction with CO and H₂ gases, as represented by Reactions (2) and (3). One pellet for each brand was non-isothermally

heated at a rate of 10 °C/min from room temperature to 900 °C in a N₂ atmosphere in a vertical thermogravimetric analysis (TGA, Rubotherm, Bochum, Germany). After the target temperature was reached, the gas composition was changed to 70 vol% N₂ and 30 vol% CO or H₂, and then the sample was kept at a flow rate of 1 L/min for 3 h. The gas composition and temperature profile defined for this test followed a standardized method [23]. The experiments were carried out twice for repeatability, and the results were averaged. The percentage degree of reduction (R_t) of the pellet was evaluated according to Eq. (4):

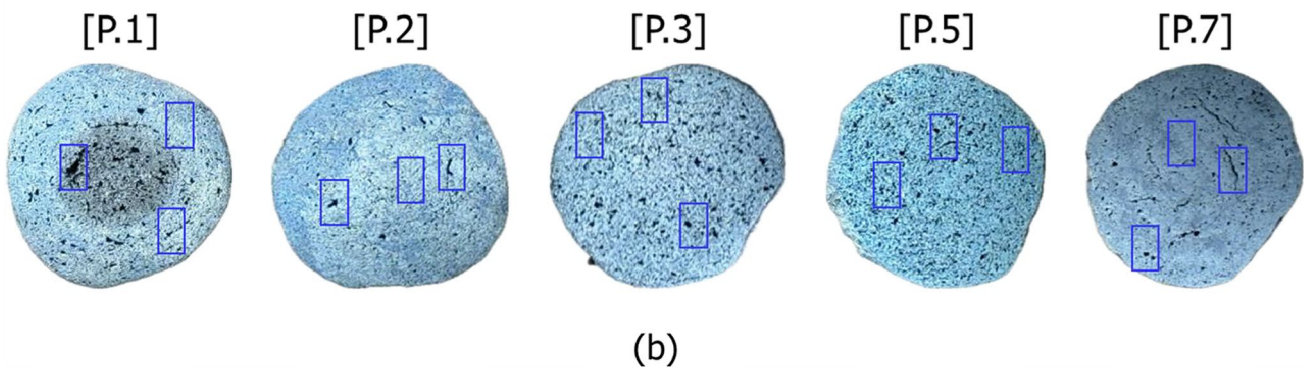
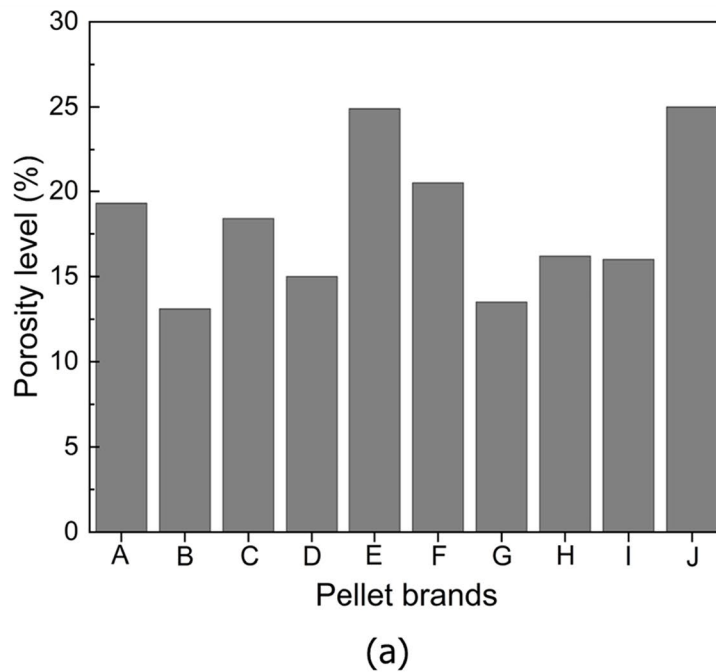
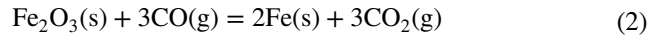
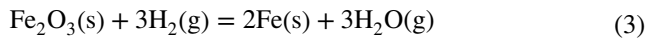


Fig. 2 **a** Porosity of several brands of iron ore pellets and **b** pore structure view by the cross-sectional images of some brands of iron ore pellets



$$R_t(\%) = \left[\frac{m_0 - m_t}{m_0 \times 0.43w_1} \times 100 \right] \times 100 \quad (4)$$

where w_1 is the T. Fe content in the pellet before reduction (wt%), m_0 is the weight of the pellet before heating (g), and m_t is the weight of the pellet after a certain reduction time (g).

The swelling behavior was evaluated by measuring the change in the pellet volume by indirect reduction with CO gas. One pellet for each brand was non-isothermally heated at a rate of 10 °C/min from room temperature to 900 °C in a N₂ atmosphere in a horizontal gold-image furnace. After the target temperature was reached, the gas composition was changed to 70 vol% N₂ and 30 vol% CO, and then the sample was kept at a flow rate of 1 L/min for 30 min. Before being charged into the reactor, each pellet had its diameter measured at three different points, and the averaged value was considered as the pellet diameter, which is then used to calculate the volume of pellet under the assumption that the pellets are perfect spheres ($\text{Volume} = \frac{4}{3}\pi r^3$). The same procedure to evaluate the diameter of pellets was applied after the experiment. The swelling index of the pellet was evaluated according to Eq. (5):

$$\text{Swelling index (\%)} = \frac{V_f - V_i}{V_i} \times 100 \quad (5)$$

where V_i is the initial volume of pellet before swelling test (mm³), and V_f is the final volume of pellet after swelling test (mm³).

Results and Discussion

Evaluation of Properties of Several Brands of Iron Ore Pellets

Ten brands of commercial iron ore pellets were analyzed for their properties such as porosity, CCS, RDI, reduction degree, and swelling index. It is necessary to understand the effect of different sources on the major properties of pellets prior to their use in the blast and/or shaft furnaces. This study focused on empirical trends mostly based on bulk chemical composition of the iron ore pellets, without deep investigation on cracks and pores shapes/sizes, phase evolution during reduction, and metallic iron morphology after experiments involving reduction procedure. However, the authors do know that the characterizations will significantly contribute to the understanding of commercial iron ore pellets behavior before being highly charged in the reactors to produce pig iron and DRI. The authors consider this study as a first chapter of a long topic of interest.

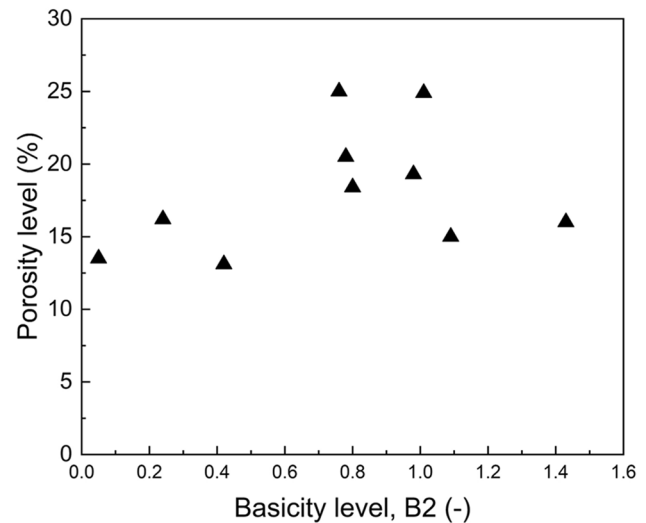


Fig. 3 Relationship between porosity and basicity (B_2)

Porosity

Figure 2a summarizes the porosity level of each iron ore pellet. According to the values shown, porosity of pellets ranged from 13.1 to 25.0%, with the lowest porosity being attributed to pellet B, while the highest level to pellet J. Two of the main factors controlling the porosity of pellets might be related to the pelletizing process, especially to the firing conditions and moisture content adopted by each pelletizing company [24]. Therefore, as these steps are very critical for the final pellet quality, it is not easy to obtain the details for the preheating and firing temperatures and time, cooling method, moisture content, and so on. However, this study tried to correlate the porosity level of pellets with an important parameter obtained by the amount of gangue components in their composition; basicity (B_2). B_2 might be considered a crucial factor, as the potential phases might be different according to their level. For example, at a lower level, the presence of Fe₂O₃ and SiO₂, in a middle level, the appearance of calcium silicates and finally, at higher levels, the formation of calcium ferrites [13, 16]. The aforementioned phases have distinct melting temperature and viscosity, which might play a determined role during the firing process, leading to less or more pores. In addition, although not deeply investigated in this study, the pore structure is a determinant factor for the final quality of the iron ore pellets, especially in terms of pore size, shape, connectivity, and so on. Figure 2b provides an overview of the difference in between the cross-sectional analysis of some pellets used in this study. The major purpose is to confirm that the pellets will have their distinct morphology at the similar porosity level. As previously mentioned, the company-specific pelletizing conditions are surely distinct, which results in

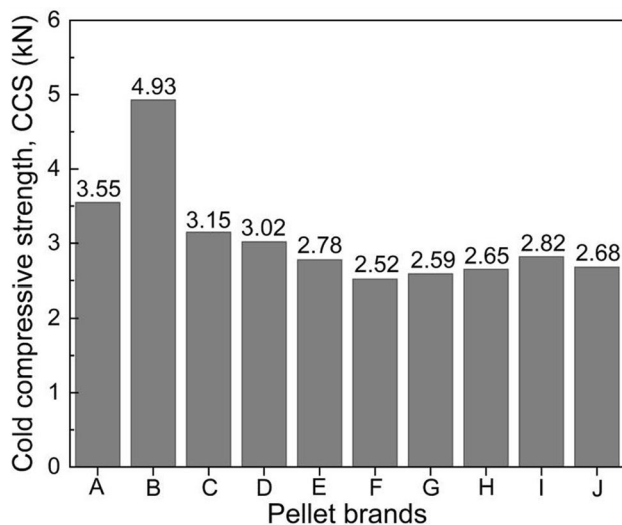


Fig. 4 Cold compressive strength (CCS) of several brands of iron ore pellets

significant change in the morphology of the pellets. Such differences might contribute to the final pellet properties even at similar porosity level. Unfortunately, such a factor cannot be further investigated. Through a detailed analysis of Fig. 2b, it is possible to differentiate the pore structure of the iron ore pellets. Pellet A showed a more concentrated porosity in the nuclei region, while pellet C and pellet E showed a more distributed porosity throughout the whole area. Pellet B presented a high densification that explains its low porosity. Pellet G, although containing a low porosity, showed an elongated and irregular shape of the pores, which might compromise its mechanical properties. In addition, the blue boxes marked at different regions represent the sample areas to be analyzed in case the pellet porosity might be evaluated by a mercury porosimeter. Since the required sampling amount is small, the results might have been distorted from the real situation. That is the reason for the selection of cross-sectional analysis for the porosity level as an alternative tool.

Figure 3 summarizes the relationship between the porosity of pellets and their basicity. In the figure, the porosity started to increase with increasing basicity around 0.5, followed by a decrease in basicity around 1.0 [16]. The lower viscosity of calcium ferrite when compared with calcium silicates might be a significant factor in reducing the amount of pores, especially during the cooling after the firing heat treatment employed to the iron ore pellets. In case of extremely acidic pellets, a dense structure could be obtained due to the major presence of Fe_2O_3 as a single phase. Similarly, Chai et al. [25] correlated the porosity of pellets with the basicity level, showing that the highest porosity was obtained in the range from 0.4 to 1.0, and then followed by a decrease. The explanation for this behavior

was through the amount of liquid phase, where the pellets with lower amounts of liquid phase showed a higher porosity, limiting the densification of the structure. Interestingly, the pelletizing conditions are also determinant factors to the final structure of pellets. A single variable, such as basicity, might affect the porosity as previously explained, in case all the other variables are constant. That is the interesting point of this study, to analyze and understand the pellets behavior for real samples, when the variables are not constant, which provides an investigation closer to the real application.

Cold Compressive Strength (CCS)

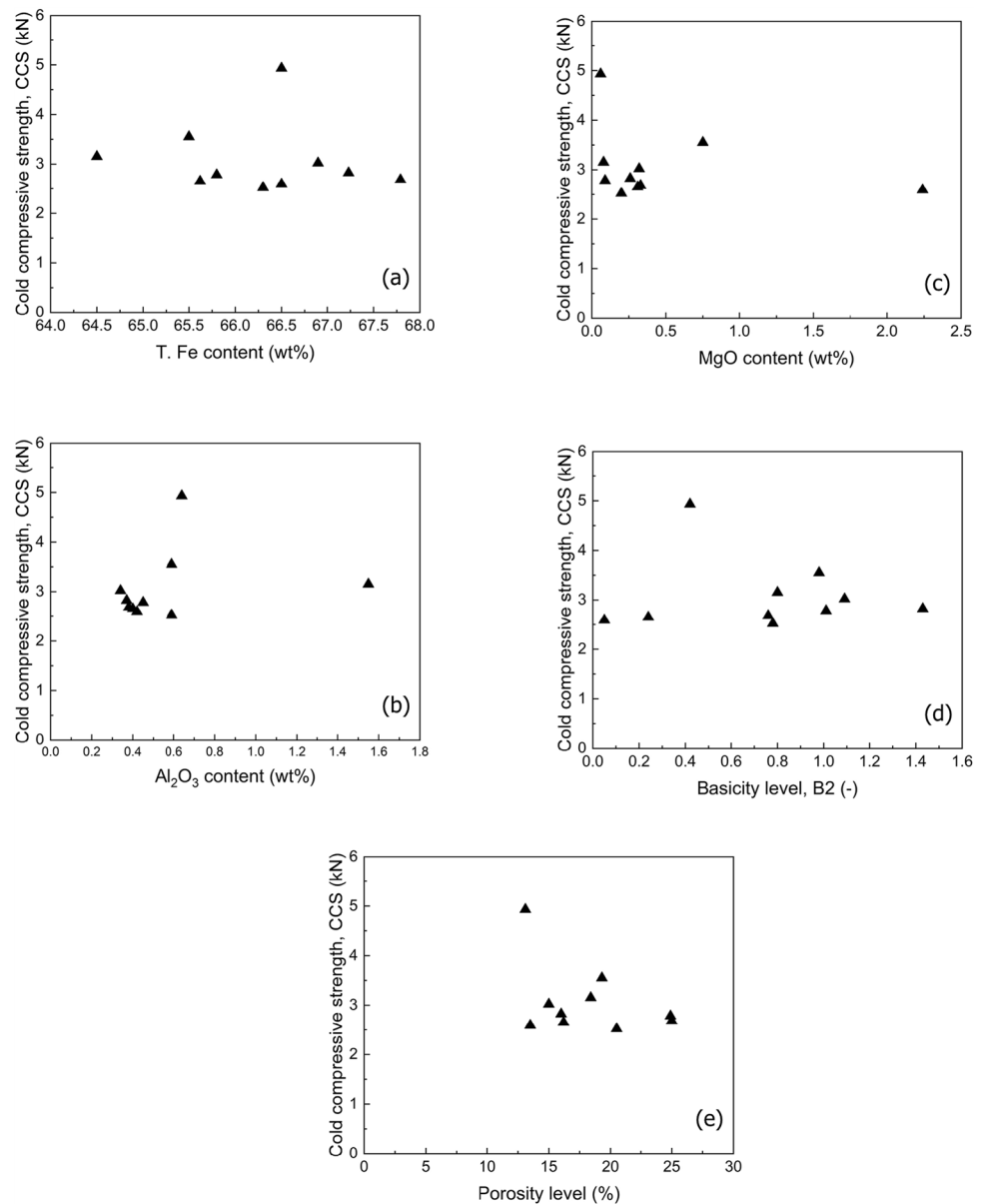
Figure 4 summarizes the cold compressive strength for the pellets. Firstly, it is necessary to mention that all brands are over the minimum requirement to be charged into blast furnaces, which is around 2 kN [26, 27]. Therefore, through the obtained results, the strongest pellet brand was found to be brand B, while the weakest was brand F. According to previous studies [13–17], several factors such as gangue components and porosity might affect the mechanical strength of iron ore pellets. That is the reason for this study to compare the results obtained in Fig. 4 for the CCS with some parameters, which includes T. Fe, Al_2O_3 and MgO contents, basicity level (B_2), and porosity level.

Figure 5 summarizes the relationship between cold compressive strength and the previously cited parameters. Each data point marked in the graphs represents an averaged value of ten pellets chosen randomly from the received package. Although the trend might not be so clear for T. Fe and Al_2O_3 contents and basicity B_2 due to the complex chemical composition of pellets, it could be concluded that lower MgO content and lower porosity are desired to achieve higher values of pellet strength. Higher mechanical strength is mostly related to the densification of the pellet microstructure. A lower porosity value promotes more densified structure, which is decisive to improve the pellet strength, while a lower MgO content promotes a higher stabilization of single-phase other than the presence of a dual-phase structure (hematite – Fe_2O_3 + magnesioferrite – $\text{MgO} \cdot \text{Fe}_2\text{O}_3$) [28, 29].

Reduction Disintegration Index (RDI)

Figure 6 summarizes $RDI_{+3.15 \text{ mm}}$ of the iron ore pellets in terms of CO and H_2 gases as reducing agents. The first notable highlight is that brand G pellet showed the worst $RDI_{+3.15 \text{ mm}}$ values independent of the reducing gas used. In the case of the best one, some changes were observed when using CO or H_2 . Brand F in the case of CO gas and brand B in case of H_2 . Moreover, the current study aimed to understand the differences in the $RDI_{+3.15 \text{ mm}}$ behavior by using both gases. Figure 7 summarizes the relationship

Fig. 5 Effect of different iron ore pellet parameters on CCS: **a** T. Fe content, **b** Al_2O_3 content, **c** MgO content, **d** basicity B_2 , and **e** porosity



between $RDI_{+3.15 \text{ mm}}$ by CO and H_2 with the reduction degree obtained after the RDI experiments. Although the linear fitting does not seem to be reliable due to low R^2 values, it is interesting to take an overlook on the data points, to point out that the $RDI_{+3.15 \text{ mm}}$ trend with reduction degree was different for both gases. As the reduction degree increased, the $RDI_{+3.15 \text{ mm}}$ by CO gas was improved while those by H_2 gas was deteriorated. A similar trend was found by Mizutani et al. [30], and the possible reason for such a behavior is that the reduction by CO gas follows a topochemical reaction, promoting a dense reduced layer in the outer shell, while the reduction by H_2 does not follow a topochemical reaction, and nucleates throughout the whole inner area of the pellet, generating significant amount of cracks.

Figure 8 summarizes the relationship between $RDI_{+3.15 \text{ mm}}$ and pellet parameters. Each data point marked in the graphs represents one test conducted with about ten pellets chosen randomly from the received package. A higher T. Fe was not desired to improve the $RDI_{+3.15 \text{ mm}}$ as more Fe_2O_3 would transform into Fe_3O_4 during the experiment, which causes significant volume expansion and consequent cracks [31, 32]. The presence of MgO in the pellet deteriorated the $RDI_{+3.15 \text{ mm}}$ value due to the stabilization of Fe_3O_4 phase as magnesioferrite ($\text{MgO} \cdot \text{Fe}_2\text{O}_3$) or because of the increase in the pore size promoted by MgO [14]. An increased basicity of pellets also deteriorated the $RDI_{+3.15 \text{ mm}}$ value as silicates are stable during reduction and when the basicity is increased, the major bonding phase is shifted from calcium silicates to calcium

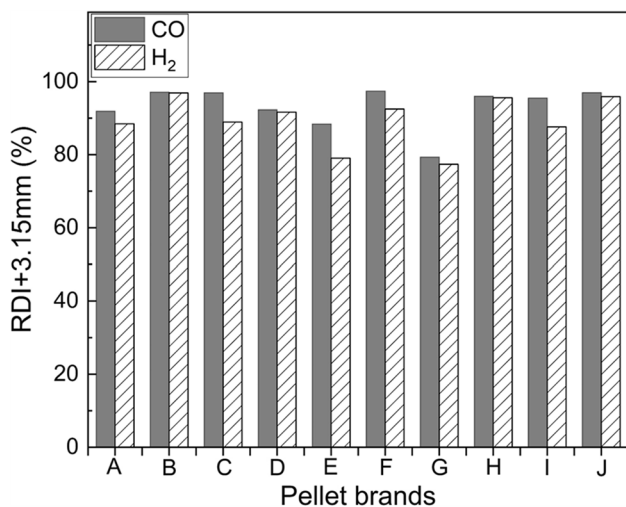


Fig. 6 Reduction disintegration index (RDI) of several brands of iron ore pellets

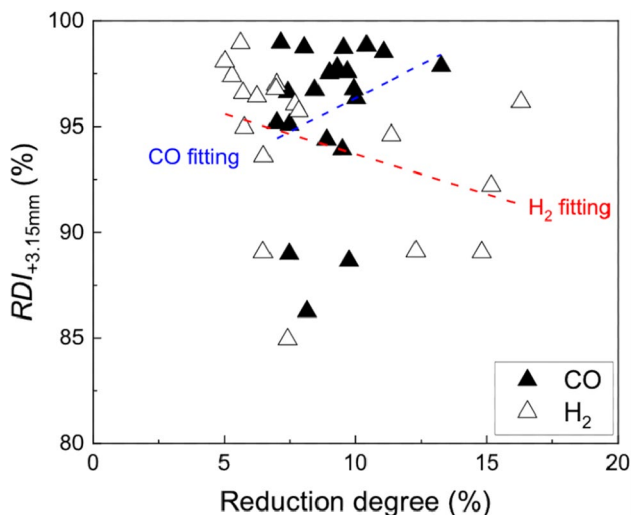


Fig. 7 Relationship between $RDI_{+3.15\text{ mm}}$ and reduction degree of pellets after RDI tests

ferrites. Then, during reduction, the hematite present in the calcium ferrites was also reduced, generating voids and cracks [13]. Al_2O_3 content does not strongly affect the RDI of the pellets. The effect of porosity follows a similar trend observed in Fig. 7, where higher porosity might lead to higher reduction degree and, consequently, a positive effect was found in the case of CO used as a reducing gas, while a negative effect resulted in the case of H_2 .

Reduction Degree (R_r)

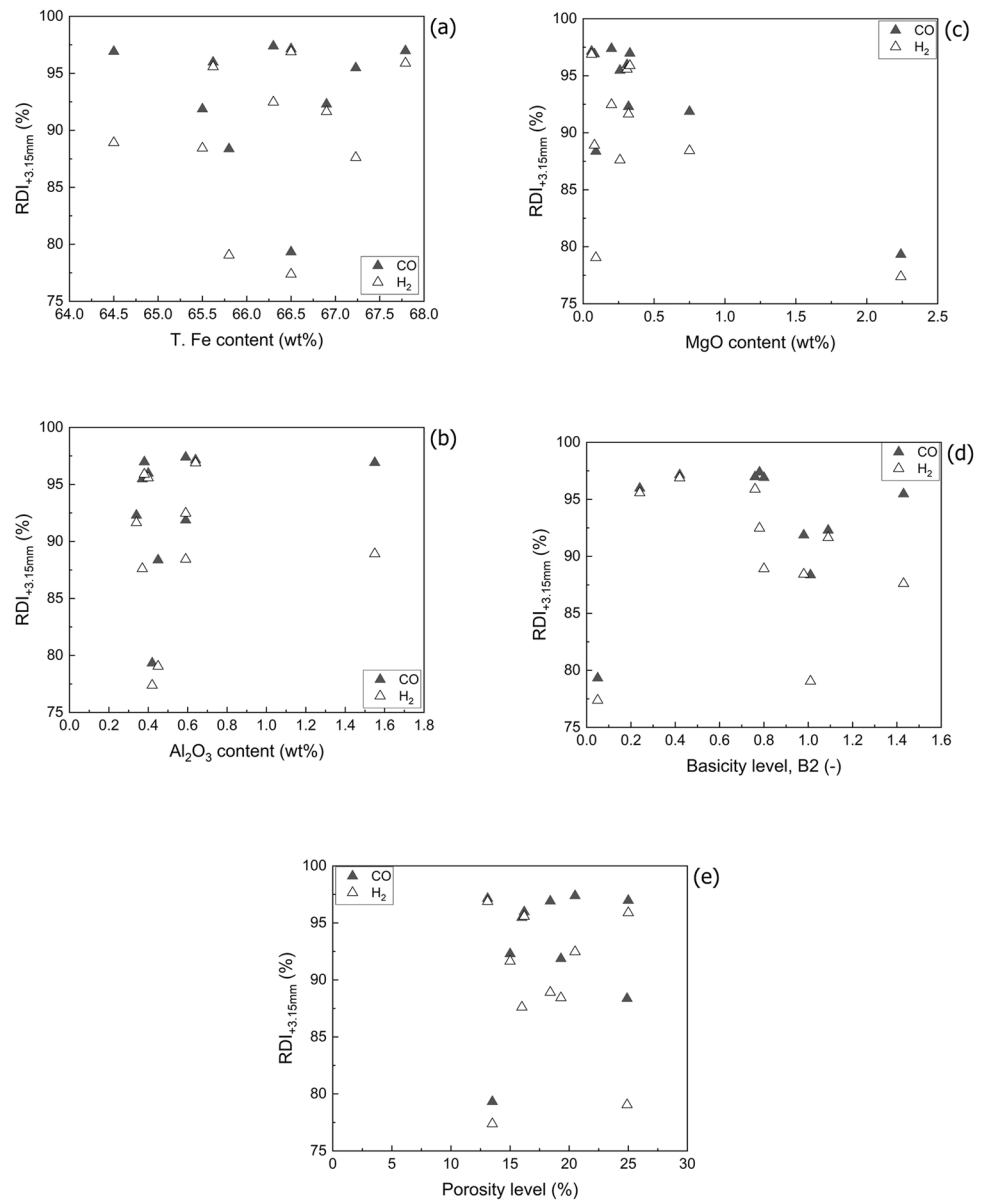
Figure 9 summarizes the reduction degree of iron ore pellet brands in terms of CO and H_2 gases as reducing agents.

The reduction behavior of pellets by CO gas showed a clear diversity in the results depending on the pellet brand, while the reduction done by H_2 had a negligible dependence on the pellet brand. As shown, the reduction degrees obtained by H_2 as reducing gas were in the similar range—all over 95%, which confirms the reducing strength of H_2 compared with CO. The highest reduction degree in terms of CO gas was obtained by brand I, while the lowest was obtained by brand B. The reduction of iron ore pellets by CO was highly affected by a combination of factors such as T. Fe (Fe_2O_3 content), Al_2O_3 and MgO content, porosity, and basicity, as shown in Fig. 10. A higher amount of T. Fe is positive for the reduction degree since a greater amount of Fe_2O_3 is present, and consequently more oxygen is available to be removed from the sample. A higher porosity level is also important for the reduction by CO since more channels for the gas to diffuse into the unreacted oxides are available. In the case of basicity, it could be explained as the calcium ferrite phase contributes to the progress in reduction, or the reducibility of this calcium ferrite phase is higher than that of silicates or single hematite [33]. That is the reason that higher basicity is preferred. The presence of MgO is not beneficial for the reduction of iron ore pellets because MgO tends to stabilize as magnesioferrite, which is an intermediate compound formed in between hematite and wustite/metallic iron [34]. As is similar to the results obtained for RDI tests, Al_2O_3 content does not play an important role in the reduction of pellets, which might be more related to the range studied. If this study fixed all the other parameters, and just varied the amount of Al_2O_3 , we might have seen some deviation provoked by Al_2O_3 . However, the current study put more focus on the understanding of the effect of different parameters for real samples, which indicates that Al_2O_3 is not an important factor affecting the pellet quality compared with other factors studied. For the results shown in Fig. 10, each data point marked in the graphs represents an averaged value of two tests conducted with one pellet each, chosen randomly from the received package.

Swelling Index

Figure 11 summarizes the swelling index of iron ore pellet brands. As shown, brand C and brand A showed the highest and lowest swelling, respectively. Figure 12a summarizes the relationship between the swelling index and reduction degree obtained after the swelling test. One of the main factors affecting the swelling behavior of pellets is the volume expansion induced by the transformation of hematite into lower valence oxides such as magnetite and wustite [32, 35]. Therefore, the reduction degree of pellet brands is an important factor to understand the swelling behavior of each pellet brand. According to the results,

Fig. 8 Effect of different iron ore pellet parameters on RDI: **a** T. Fe content, **b** Al_2O_3 content, **c** MgO content, **d** basicity B_2 , and **e** porosity



a higher reduction degree resulted in higher swelling, which is a harmful consequence for the strength of pellets. To exemplify, Fig. 12b correlated the swelling index of pellets with the CCS of pellets after the swelling test. A higher swelling ratio deteriorated the pellet strength, which is not desirable for the blast furnace operation. The maintenance of pellet structure and strength is of great importance for the smooth operation of the reactor. The porosity of pellets is also a contributing factor for the swelling. As the swelling is related to the volume expansion, a higher porosity is desired for a lower swelling to properly accommodate the change from the volume expansion. In the case of a lower porosity, the changes might not find enough space inside the actual pellet, promoting the formation of cracks and higher swelling. In

addition to the effect of porosity, Fig. 13 summarizes the effect of T. Fe, Al_2O_3 , MgO content, and basicity of pellet on the swelling behavior of pellets. The effect of T. Fe content seems to be complex, which might be determined by other factors. The starting idea is that a high T. Fe content is detrimental to the swelling of pellets as more Fe_2O_3 is available to further reduce into lower valence oxides such as Fe_3O_4 and FeO , which would result in larger volume expansion. The aforementioned phenomenon is observed by the effect of Al_2O_3 , which is a Fe_2O_3 stabilizer. Higher levels of basicity and porosity improve the swelling of pellets by different reasons. Higher basicity means that a significant amount of Fe_2O_3 is in the form of $\text{CaO}\cdot\text{Fe}_2\text{O}_3$, which reduces the volume expansion caused by the reduction of single hematite, and the

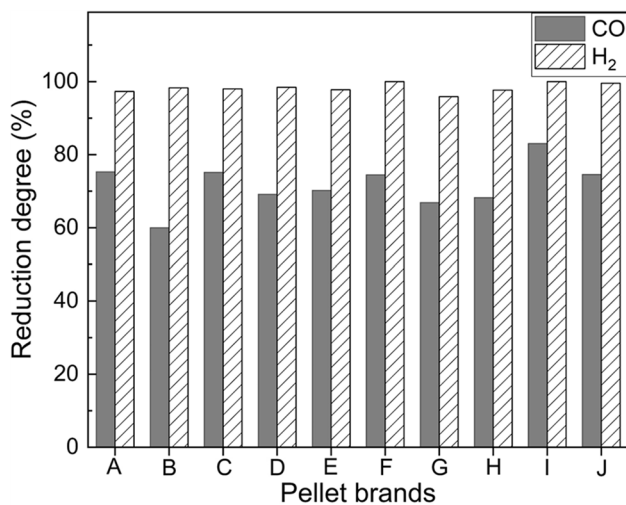


Fig. 9 Reduction degree of several brands of iron ore pellets

difference in the reduction rate of $\text{CaO} \cdot \text{Fe}_2\text{O}_3$ and calcium silicates. In the case of porosity, higher levels might be favorable to accommodate the volume expansion caused by the reduction. For the results shown in Fig. 13, each data point marked in the graphs represents an averaged value of two tests conducted with one pellet each, chosen randomly from the received package.

Effect of Pellet Size on the Evaluated Properties of Iron Ore Pellets

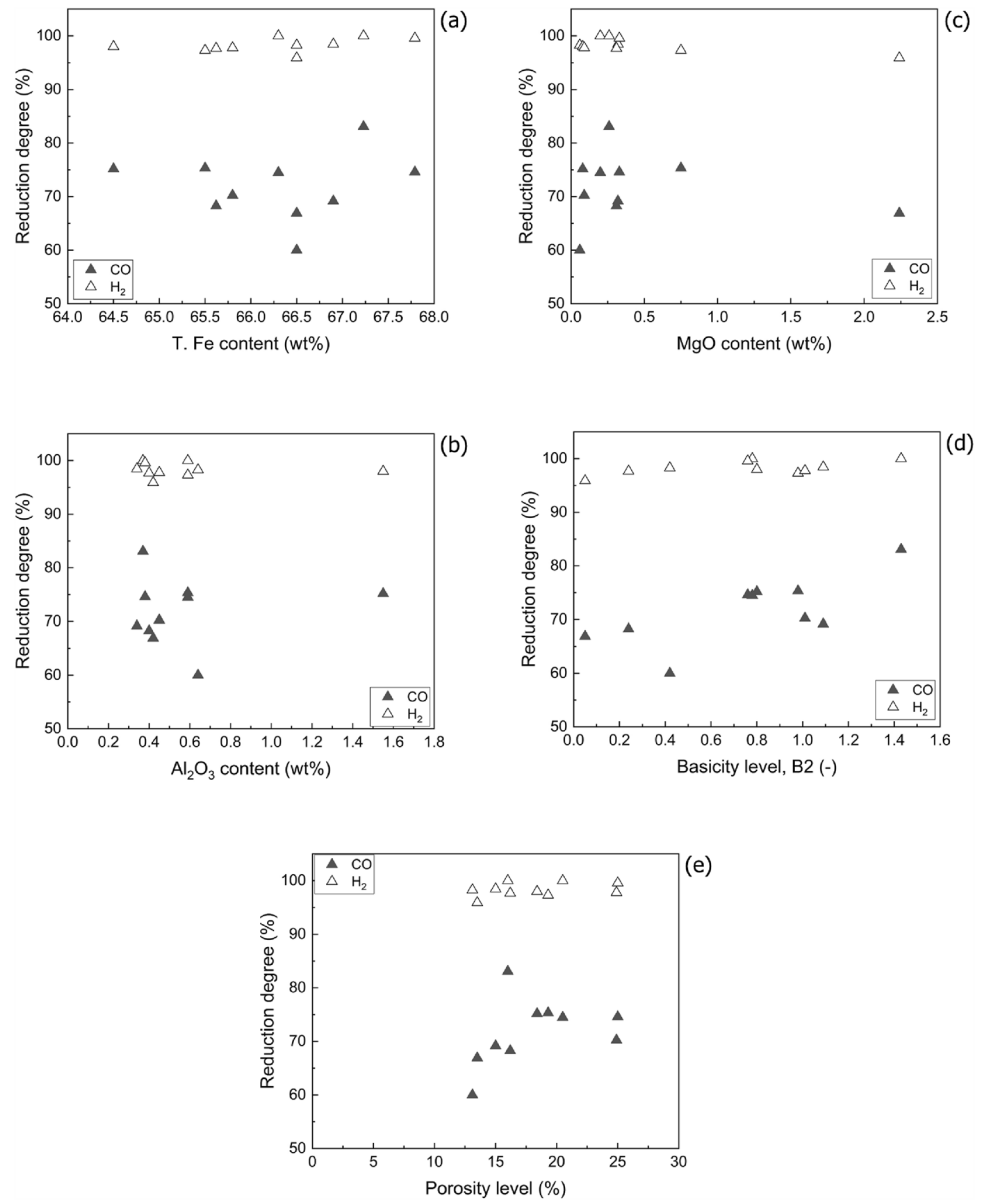
To investigate the effect of pellet size on the previously analyzed properties, three different brands were selected: brands H, I and J. Iron ore pellets have a diverse range of size, mostly ranging from 8 to 16 mm [24]. Therefore, it is important to understand how the pellet size affects each property of the pellet, and to determine selection criteria of which size should be used in the actual operations. This study divided the pellets into four size groups: group 1 from 8 to 9.5 mm, group 2 from 9.5 to 11 mm, group 3 from 11 to 13 mm, and group 4 from 13 to 16 mm. All groups were analyzed in the identical experimental conditions. Figure 14 summarizes the effect of pellet size on CCS, $RDI_{+3.15 \text{ mm}}$, reduction degree, and swelling. Based on the results, the trends were very clear. A bigger pellet improved the compressive strength and swelling behavior of pellets but deteriorated the $RDI_{+3.15 \text{ mm}}$ and reducibility. The relationships presented for $RDI_{+3.15 \text{ mm}}$, reduction degree, and swelling are in agreement with previous researches [36, 37]. In case of CCS, some reported works provided an opposite relationship than the one found in our current investigation. Therefore, the same trend agrees with previous research reported by Iljana et al. [38] The possible reason for a lower compressive strength by

increasing the pellet size is the presence of a dualistic structure represented by an outer shell composed by hematite and a nucleus composed of magnetite, which could not be completely oxidized during the pelletizing process. However, through the chemical analysis of the received pellets, there was no significant presence of unoxidized magnetite in the core of studied pellets, which might eliminate this reason to explain the effect of pellet size on CCS. In the case of RDI, reducibility and swelling, the explanation comes from the same reason since all these properties are directly related to the reduction degree of the pellets. Smaller pellets are reduced more as the available oxygen is lower for the same flow rate of reducing gas. This behavior formed a dense reduced shell, improving $RDI_{+3.15 \text{ mm}}$, and a higher volume expansion due to the transformation of Fe_2O_3 into Fe_3O_4 , resulting in a higher swelling ratio. Therefore, based on the obtained results, an intermediate size of pellets (10 to 14 mm) might be the best choice for a smooth operation without compromising both strength and reduction behavior of pellets.

Multiple Linear Regression (MLR) Analysis

Several experiments were conducted using ten kinds of iron ore pellet brands, commonly used by the major steelmaking companies in the world, to first map the difference in their physicochemical properties, and then to understand how the distinct chemical composition and morphology—in terms of porosity level—would affect their physicochemical properties. This study represented the T. Fe (Fe_2O_3), Al_2O_3 and MgO content as an effect of chemical composition, and the basicity B_2 , which are the most decisive components of an iron ore pellet—over 98% according to Table 1. Numerous experiments were performed to carefully clarify the effect of each parameter on the physicochemical properties of pellets; this study attempted to derive an equation for each property, including CCS, $RDI_{+3.15 \text{ mm}}$, reduction degree, and swelling. To facilitate the interpretation of the experimental results, multiple linear regression (MLR) analysis was employed as an exploratory statistical tool to identify empirical relationships between pellet characteristics (Fe_2O_3 , Al_2O_3 , MgO, basicity B_2 , and porosity) and the measured pellet properties (CCS, RDI, reduction degree, and swelling index). Because the investigated dataset consists of a limited number of commercial pellet brands with inherently correlated variables, the resulting models should not be regarded as mechanistic or universal predictive equations. Instead, the MLR analysis was used to estimate the relative influence of the investigated parameters within the present dataset and to support the interpretation of the observed experimental trends. Online supplementary Table S1 summarizes the derived relations in between each property and major pellet compositional parameters, while more detailed statistical information is

Fig. 10 Effect of different iron ore pellet parameters on reduction degree: **a** T. Fe content, **b** Al_2O_3 content, **c** MgO content, **d** basicity B_2 , and **e** porosity



expressed in online supplementary Tables S2–S6. The mentioned tables can be found in the online supplementary file.

It is important to mention that in the case of swelling, a higher value is not desirable, which indicates that the pellet swells more, and consequently is weaker after reduction. From online supplementary Table S1, the linear regression coefficient (R^2) was also reported for the derived equations. The authors do know that the R^2 value is not high enough to provide strong reliability of the statistical models presented. However, although the linearity does not look high for some cases, especially that for CCS, it should be highlighted that this study considered the simultaneous effect of several parameters by using several commercial brands of pellets, which means that in case one parameter had its effect evaluated, other

parameters could not be maintained constant, deteriorating the relationship of the analyzed variable with the correspondent property. Therefore, such an approach improved the current study using real samples, not modified samples, to understand the effect of a single parameter on the pellet properties, which was mostly reported in the literature [13–17, 29]. To provide a more transparent and trustful model, it is for sure necessary to increase the amount of samples, but the authors doubt that the availability of commercial pellets would increase this list to a more significant amount. As mentioned, the received brands are the most used pellets in the major blast furnaces being operated worldwide. Therefore, owing to the limited sample amount and interdependence among variables, the results should be interpreted as empirical trends

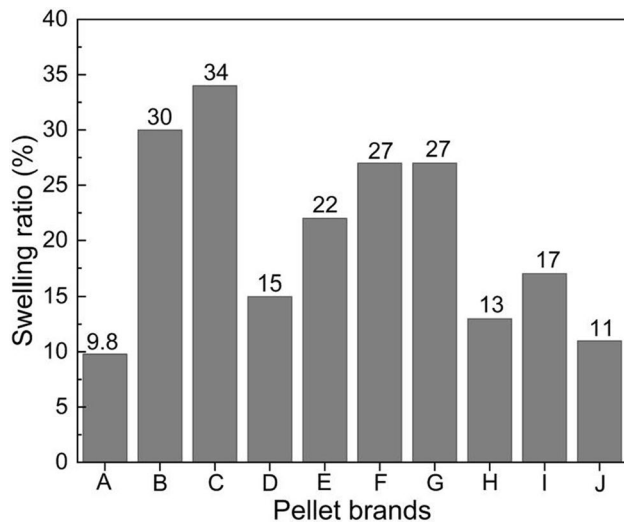


Fig. 11 Swelling of several brands of iron ore pellets

rather than independent mechanistic effects. It is necessary to highlight that the analyses were not conducted for the degree of reduction by H_2 gas as the reducing agent since the results were almost independent of pellets parameters. The authors do know that pellets might behave differently inside the reactor when H_2 is used, but this study is assessing the reduction behavior of pellets under the same conditions of hydrogen reduction, and based on the results, all the brands achieved a very high reduction degree independent of their chemical composition and porosity. As shown in Fig. 9, the reduction degree was achieved above 95% for all the pellets, which indicates that the reducing strength of H_2 is more influential than the pellet quality. Furthermore, the current study aims to clarify the individual effect of each variable on each property to give a better understanding of the governing factors for a selection criterion of pellets to be used especially in the blast furnace, which includes the optimal range of each variable. According to Table 1, the T. Fe content of all brands used in this study were quite similar since the manufacturing of iron ore pellets always considers the use of high-grade ores as raw materials. However, high-grade ores will be depleted in coming years, thus it is possible to use lower-grade iron ore for the pelletizing process, which will provide a wider range of T. Fe content. Therefore, this study did not perform independent analysis by T. Fe content.

To facilitate comparison among the investigated variables, the standardized regression coefficients obtained from the MLR analysis were normalized and summarized in a heatmap chart, as shown in Fig. 15. Since the coefficients are dimensionless, their magnitude provides a direct indication of the relative influence of each parameter on the corresponding pellet property. It should be emphasized

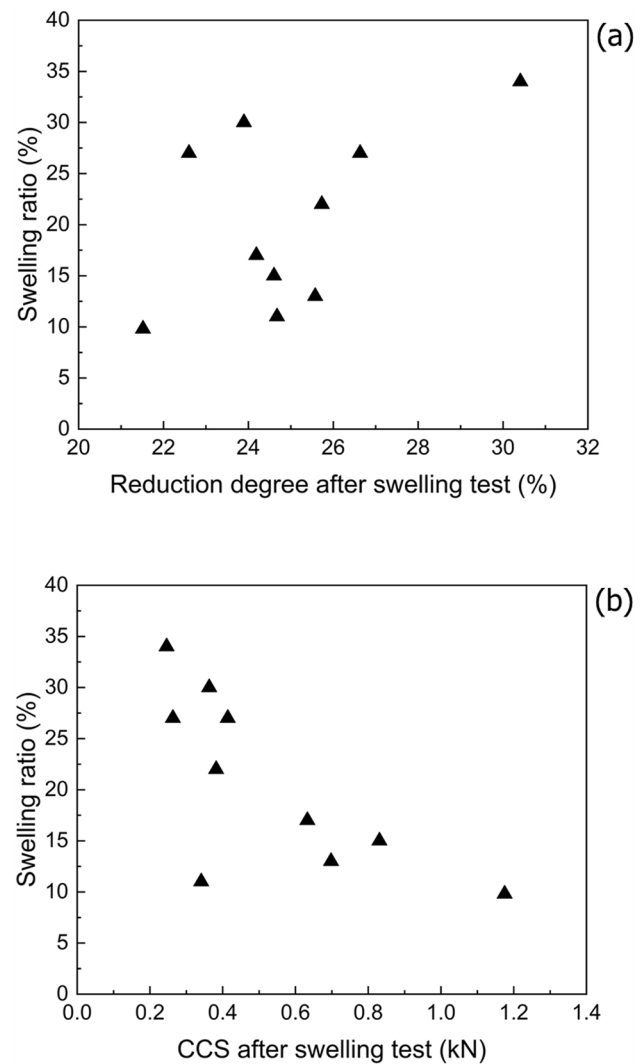
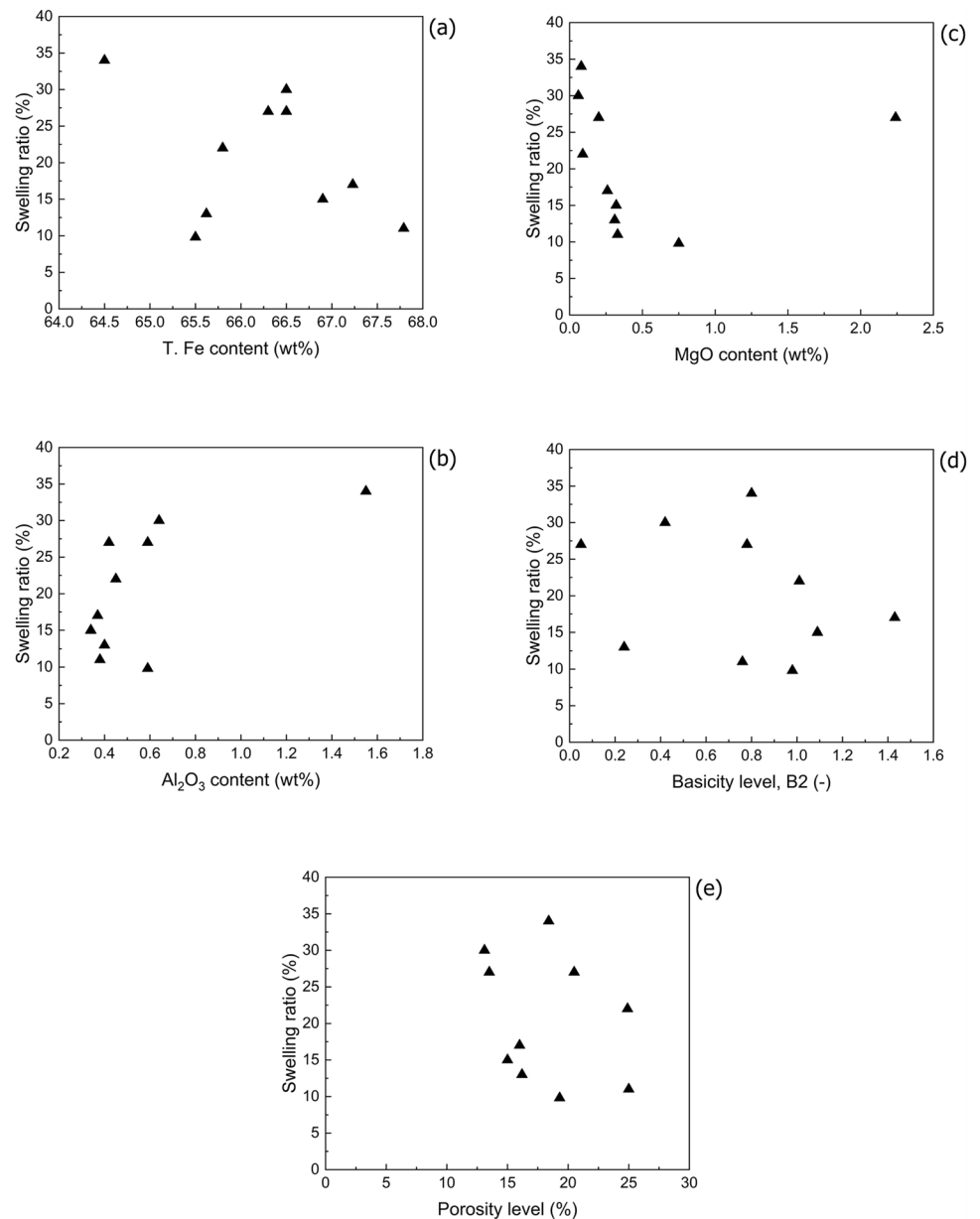


Fig. 12 Relationship between swelling and **a** reduction degree of iron ore pellets after the swelling tests and **b** CCS of pellets after the swelling tests

that these coefficients represent empirical trends within the investigated commercial pellet dataset and should not be interpreted as independent mechanistic effects. The values were normalized by the maximum absolute standardized coefficient observed among all models. Therefore, the chart illustrates the relative importance of each variable rather than the magnitude of the original regression coefficients.

Figure 15 summarizes the relative influence of the investigated pellet characteristics on CCS, RDI, reduction degree, and swelling index using normalized absolute standardized regression coefficients obtained from the MLR analysis. The normalization was performed with respect to the maximum coefficient observed among all models, allowing direct comparison of the relative contribution of each variable. The results indicate that MgO and basicity generally exhibited the strongest associations with

Fig. 13 Effect of different iron ore pellet parameters on swelling: **a** T. Fe content, **b** Al_2O_3 content, **c** MgO content, **d** basicity B_2 , and **e** porosity



pellet performance, while porosity and Al_2O_3 showed a comparatively smaller influence. In addition, the dominant variables varied depending on the evaluated property, suggesting that no single pellet characteristic can adequately describe overall pellet quality. These observations suggest that structural characteristics, represented by pores structure and mineral phase constitution, may play a more significant role than minor compositional variations in determining the overall performance of commercial pellets. However, because the regression models were developed from a limited set of commercial pellets with interdependent variables, these relationships should be interpreted as empirical trends rather than mechanistic or universally predictive effects.

The predictive capability of the empirical MLR models was evaluated by comparing the measured experimental values with those calculated from the regression equations, as shown in Fig. 16. In general, the calculated values followed the observed experimental trends, indicating that the selected compositional and structural parameters capture a substantial portion of the variability among the investigated pellet brands. Nevertheless, owing to the limited number of commercial samples and the interdependence among variables, the models should be interpreted as exploratory statistical tools rather than universally applicable predictive equations. The purpose of this comparison is not to establish a predictive design tool, but rather to assess the consistency of the empirical relationships identified in the present study.

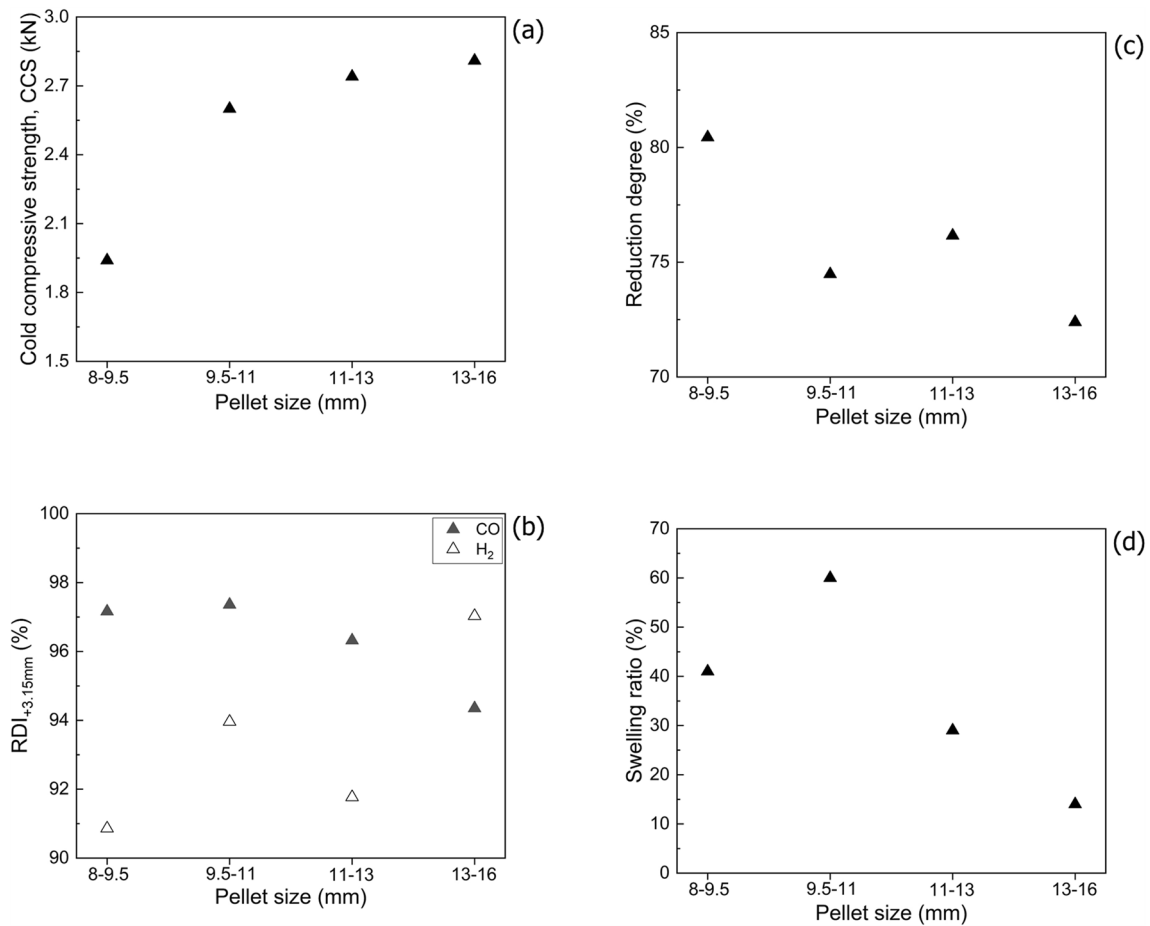


Fig. 14 Effect of iron ore pellet size on its performance: **a** CCS, **b** $RDI_{+3.15\text{ mm}}$, **c** reduction degree, and **d** swelling

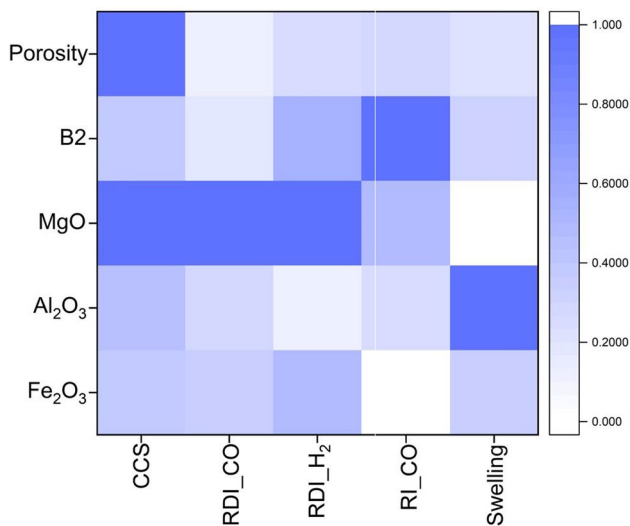


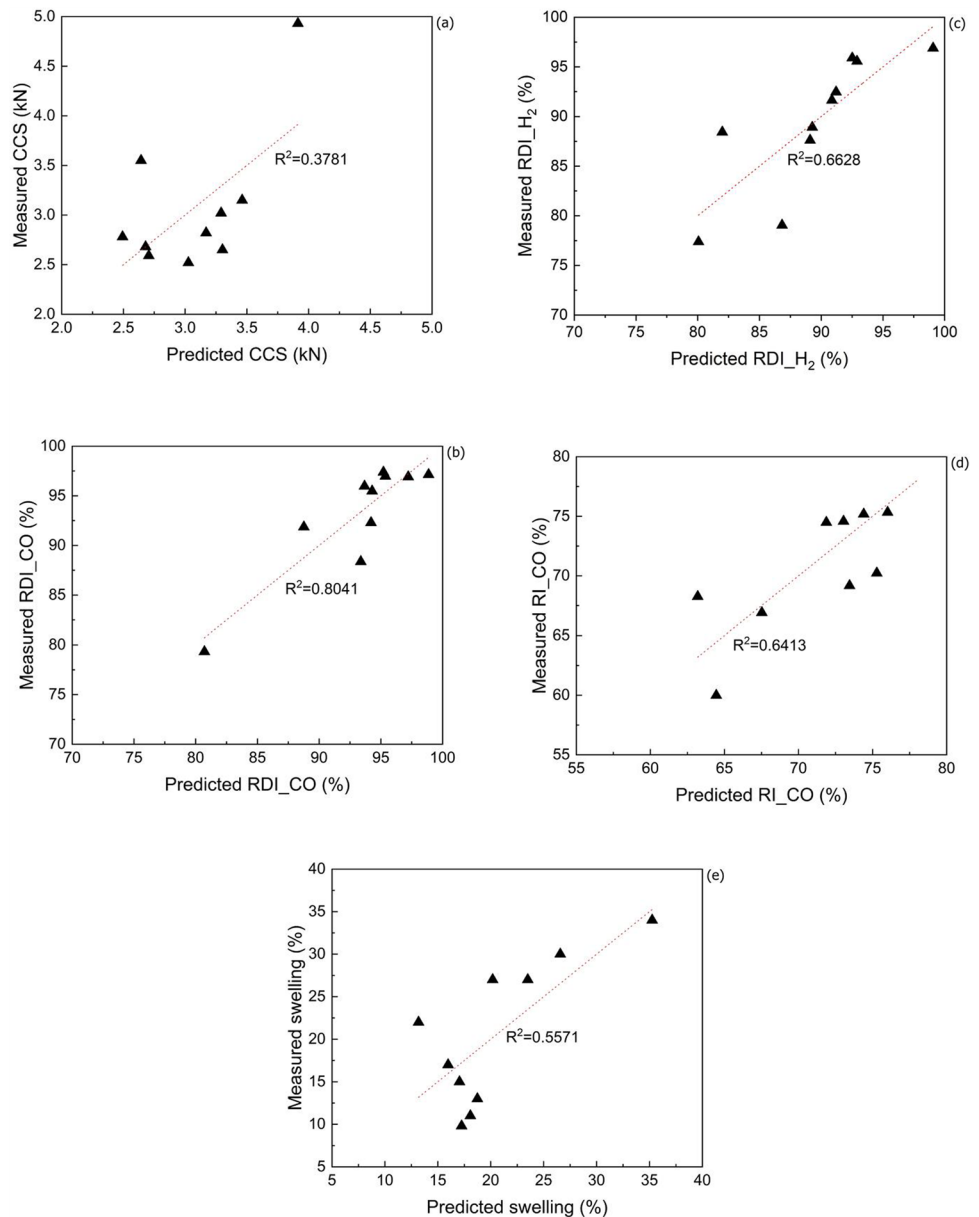
Fig. 15 Heatmap of normalized absolute standardized regression coefficients ($|\beta|$) obtained from the MLR analysis, illustrating the relative influence of pellet characteristics (Fe₂O₃, Al₂O₃, MgO, B₂, and porosity) on CCS, RDI, reduction degree, and swelling behavior. Darker colors indicate stronger relative influence

Relative Influence of Iron Ore Pellets Parameters on Their Physicochemical Properties

According to the trend observed in this study regarding the effect of basicity on the physicochemical properties of iron ore pellets, an intermediate level of basicity might be a proper option to maintain all properties at acceptable levels. Higher basicity leads to good reducibility and swelling, while intermediate levels of B_2 improved the compressive strength and reduction disintegration index. A similar finding was reported by Umadevi et al. [13], which suggested an optimal basicity level between 0.3 and 0.4 to satisfy all the measured properties such as CCS, TI, and RDI. Therefore, after deep investigation on the effect of basicity level on pellet performance, B_2 basicity around 1.0 might be the optimal choice for a better optimization of the pellet quality.

In the case of Al₂O₃ content, its individual effect is still unknown, and there is lack of data regarding this issue, which might be related to the most use of high-grade iron ores as raw materials for the pelletizing process, excluding high Al₂O₃ ores. Therefore, the utilization of such iron ores was

Fig. 16 Comparison between experimentally measured and MLR-calculated values for the investigated pellet properties: **a** CCS, **b** RDI +3.15 mm by CO, **c** RDI +3.15 mm by H₂, **d** reduction degree by CO and **e** swelling. The coefficient of determination (R^2) is provided for reference. The models are intended to describe empirical trends within the investigated commercial pellet dataset rather than to provide universal predictive equations



not studied in the manufacturing of pellets. However, several investigations are being conducted for the use of high- Al_2O_3 ore for the sintering process [34, 39], but the study for the pelletizing process is still ongoing. Agrawal et al. [17] analyzed the effect of Al_2O_3 on the CCS and RDI of pellets, pointing out that the increase in Al_2O_3 content deteriorates both properties. They considered the Al_2O_3 content in between 1.7 and 2.7 wt%, which is much higher than the common Al_2O_3 content being used in current pelletizing processes. Among the ten brands used in this study, only one (brand C) showed an Al_2O_3 content over 1 wt%. The results obtained from Agrawal et al. [17] might be originated from the iron ore source used for the pelletizing process, which required the change in the pelletizing conditions. A similar investigation

was also conducted by Umadevi et al. [40], which studied two different Al_2O_3 -containing ores as raw materials for pelletizing with contents above 2 wt%. Through the results, the final quality of the pellet seems the same. Samples constituted of majorly hematite, followed by porosity and silicate melt, similar values of CCS and RDI. Zhang et al. [41] studied the effect of aluminum oxide on the CCS of pellets with Al_2O_3 contents ranging from 1 to 3.5 wt%. According to his findings, CCS was improved from 1 to 1.5 wt%, followed by a deterioration. The major reason for the early improvement is that when Al_2O_3 is present in small levels, it will enter the silicate system to form a compound ($2\text{FeO} \cdot 2\text{Al}_2\text{O}_3 \cdot 5\text{SiO}_2$), which will reduce the amount of iron combined with SiO_2 that hinders the oxidation of Fe_3O_4 into Fe_2O_3 , improving

the crystallinity of Fe_2O_3 . Therefore, the individual effect of Al_2O_3 is still unclear, and the results obtained in this study might be helpful to understand the Al_2O_3 behavior in the pellets. This study recommends, based on the experimental results, an Al_2O_3 content below 1 wt% for the pellets to be abundantly charged into the reduction furnaces. However, if other parameters are modified to compensate for the increase in Al_2O_3 content with porosity or basicity level, then additional investigation should be conducted to adjust the optimal parameters for each Al_2O_3 content. While such research is not provided, levels below 1 wt% might be applied.

As is similar to the Al_2O_3 content, few studies were done to clarify the effect of MgO content on the pellet quality. However, MgO is known for its high melting point, mostly used as refractory or slagging material. In particular, MgO behavior in the sintering process is widely known to stabilize the $\text{MgO} \cdot \text{Fe}_2\text{O}_3$ phase, which has lower reducibility than other phases containing Fe_2O_3 , deteriorating RDI and swelling due to the similar reason. Therefore, the role of MgO in iron ores in the high temperature processes, either sintering or pelletizing, is well-known. A similar conclusion was obtained in this study, that is, the increase in MgO content deteriorated all analyzed properties, which agrees with some previous studies [13, 29]. Such conclusions confirm that a low MgO content of around 0.2 wt% is preferable for the pellets to be highly used in the reduction furnaces. Although one successful case of implementing high-MgO pellets in blast furnace is known [42], several previous studies corroborate our findings, that the high presence of MgO deteriorates some of the physicochemical properties of pellets. Various reasons could be mentioned, for example, the pelletizing process adopted by different companies when treating high-MgO ores, which is a valuable step and determines the final quality of the pellet in terms of complete firing, porosity level, etc.

As is similar to the effect of basicity, the porosity also has dual effects on the properties of iron ore pellets. An increase in porosity leads to loose structure, and deteriorates the mechanical strength [13], while promoting the reduction. Simultaneously, this loose structure provides enough space to accommodate any volume expansion that occurred from the reduction process. Therefore, intermediate levels of porosity are also preferred to maintain all the investigated properties at acceptable levels. This study investigated the range of porosity from around 15 to 25%, which results in selecting a porosity level at about 20% to keep the pellet quality in a safe zone for the measured properties.

Conclusions

Several analyses were performed in this study to understand the effect of some pellet parameters on the physicochemical properties of iron ore pellets such as CCS,

$\text{RDI}_{+3.15 \text{ mm}}$, reducibility, and swelling ratio. The major parameters investigated in this study were T. Fe content, basicity level, Al_2O_3 content, MgO content, and porosity level. From the findings, the following conclusions were obtained.

- (1) The pellet parameters distinctly affect several properties investigated in this study. The pellets might be selected on the basis of their chemical composition and mechanical strength, mostly measured by CCS. Based on these criteria, brand B pellet might be considered the first option for steelmaking companies. However, careful investigation indicated that the pellet was analyzed to have the lowest reducibility and highest swelling, which means that the selection cannot be only based on chemical composition and mechanical strength. The pellet behavior in the blast furnace must be considered an important factor to determine what kind of raw materials should be used.
- (2) Multiple linear regression (MLR) analysis was used as an exploratory statistical approach to estimate the relative influence of pellet characteristics within the investigated commercial dataset. The equations correlated four kinds of properties such as CCS, RDI, reducibility, and swelling with several variables from the pellets, including Fe_2O_3 , Al_2O_3 and MgO content, basicity B_2 , and porosity.

$$\text{CCS: } 0.10 \times [\text{Fe}_2\text{O}_3] - 0.35 \times B_2 + 0.47 \times [\text{Al}_2\text{O}_3] - 0.55 \times [\text{MgO}] - 0.09 \times \text{porosity} - 4.66$$

(3)

$$\begin{aligned} \text{RDI}_{\text{CO}}: & 1.35 \times [\text{Fe}_2\text{O}_3] - 2.54 \\ & \times B_2 + 4.43 \times [\text{Al}_2\text{O}_3] - 8.29 \\ & \times [\text{MgO}] - 0.13 \times \text{porosity} - 28.69 \end{aligned}$$

$$\begin{aligned} \text{RDI}_{\text{H}_2}: & 2.26 \times [\text{Fe}_2\text{O}_3] - 8.31 \times B_2 + 2.20 \\ & \times [\text{Al}_2\text{O}_3] - 9.84 \times [\text{MgO}] - 0.37 \times \text{porosity} - 108.37 \end{aligned}$$

$$\begin{aligned} \text{Reduction degree}_{\text{CO}}: & 0.08 \times [\text{Fe}_2\text{O}_3] + 12.57 \times B_2 \\ & + 3.81 \times [\text{Al}_2\text{O}_3] + 3.87 \times [\text{MgO}] + 0.35 \times \text{porosity} + 44.77 \end{aligned}$$

$$\begin{aligned} \text{Swelling : } & 1.83 \times [\text{Fe}_2\text{O}_3] - 5.70 \times B_2 \\ & + 19.86 \times [\text{Al}_2\text{O}_3] - 0.30 \times [\text{MgO}] \\ & - 0.37 \times \text{porosity} - 152.50 \end{aligned}$$

- (4) Considering the charge of iron ore pellets into different reduction furnaces, within the investigated commercial pellet dataset, favorable performance was generally associated with lower MgO content, moderate basicity, and porosity near 20%. These observations should be regarded as empirical trends rather than universal design criteria.

- (5) Larger pellets improved the compressive strength and swelling, while smaller pellets enhanced the $RDI_{+3.15\text{ mm}}$ and reducibility. Therefore, intermediate size of pellets ranging from 9.5 to 12 mm is desired to maintain their performance in the reactors.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s40831-026-01608-7>.

Conflict of Interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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